



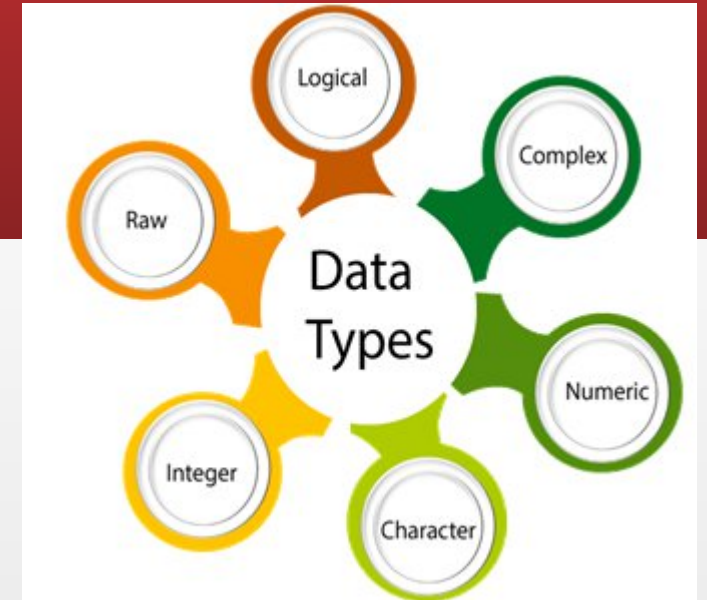
thuRsdays

LECTURE 2- CLICK IT, CODE IT
OR BOTH - THERE R CHOICES



Recap

- **What is R**
 - Statistical and mathematical programming environment
- **Origins and Evolution of R**
 - Started as a statistical analysis tool
 - Has evolved to something much bigger
- **R is now a comprehensive Data Analytic Platform**
 - Still shines with statistical analysis
 - GIS, Time–Series, Image Processing, Audio, Text mining
- **Using R requires some knowledge of Data Types and Structures**



Using R – Command Line

- R can be run from the Command Line
 - Shells such as BASH or Terminal in Mac and Windows
 - Windows terminal or power shell
 - Useful for running simple commands
- R Script is a text file that stores a sequence of commands
 - Can write it using any text editor
 - Notepad, Kate, TextEdit

Command Line is the best options for Power Users and Highly Memory Intensive tasks

No overhead of having an extra program to Run R

Not very intuitive.

R using a GUI

- GUI is a graphical user interface
 - Allows R functionality through menus and buttons
- R ships with a native R GUI for Windows and Mac
 - Installed when R is installed on your computer
 - RGui.exe in windows and R.app in MacOS
 - Allows you to write scripts and execute them, add libraries, etc.
- R GUI helps running R in one place
 - Does not offer any R functions as menus or Buttons

If you use R GUI you still need to know R Commands and Functions

There is no help with the syntax other than what is provided via native R

R via IDE

- Interactive Development Environment (IDE) is much more than a GUI
 - Like a GUI puts scripts, console and execution tools all in one place
- Gives you tools to debug your programs
 - Set watchpoints
 - Provide autofill and other context sensitive help
 - Plots are integrated within the IDE
- RStudio is a popular IDE to run R
 - You need to install R before installing RStudio
- Several other IDE options
 - VSCode, Jupyter Lab

While IDE's offer interactive help, they do not provide menus and buttons to perform analysis

You will still have to write R Code in a script or console to run it

Better than RGui because of the help that is available

The interface is more polished

GUI Based R analysis Tools

- These tools are different than GUI and IDEs
- The functionality of R is provided as a menu or a button (GUI)
 - Exploratory Data Analysis
 - Hypothesis Tests
 - Inferential Statistics
 - Regression and model fitting
 - Predictive Statistics
- You don't need to know the R programming syntax
 - The functional GUI will call R on your behalf to perform the analysis
 - Very good for beginners and those transitioning from GUI based analysis systems
 - SPSS, STATA, MINITAB

GUI-based R Analysis tools are easier to use, but only cover a small sample of R capabilities

GUI Based R Analysis Tools

- There basically two types of such tools
 - Code Exposing GUI
 - Code Hiding GUI
- Tools that are added to R as a package
 - RCommander
 - Can use GUI
 - GUI calls R functions and exposes it
 - Great for learning R syntax
- Code Hiding GUI tools
 - Jamovi is an example
 - Uses R internally
 - You don't see the R Code or R interface
 - Polished Menus

RKward is another R based GUI that can expose Code under certain conditions.

Also based on R – Comes pre-packaged or can be used with you R installation

Jamovi

- Free to download
- <https://www.jamovi.org/index.html>
- Available on major platforms
- Very polished interface
- Uses R in the background



FeaturesCloudDownloadAboutResources

Use Cloud free

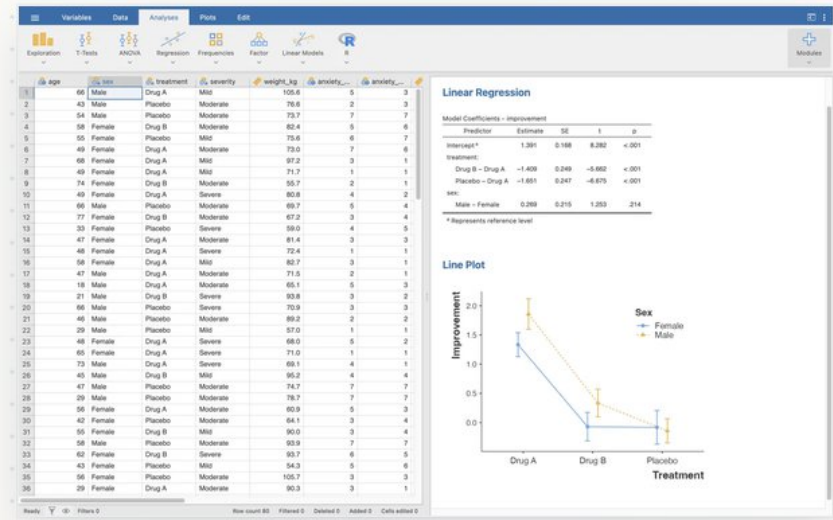
Free & Open Source

Stats made simple.

jamovi is a free and open statistical spreadsheet. Designed from the ground up to be easy to use, it's a compelling alternative to costly proprietary stats products.

Use Cloud freeDownload Desktop

Desktop is always free. Cloud's Guest plan is free too: no install, no credit card.



The screenshot displays the Jamovi software interface. On the left, a data table is visible with columns for age, sex, treatment, severity, weight_kg, and anxiety. The 'treatment' column includes categories like Drug A, Drug B, and Placebo. On the right, a 'Linear Regression' analysis is shown, with a table of model coefficients and a line plot titled 'Line Plot' showing 'Improvement' on the y-axis and 'Treatment' on the x-axis. The line plot compares improvement for Female (blue line) and Male (yellow line) across Drug A, Drug B, and Placebo treatments.

BY THE NUMBERS

Used by researchers, educators, and students worldwide.

5M+
DOWNLOADS

300+
UNIVERSITIES

70+
LIBRARY MODULES

RKward

- Offers both GUI and R Coding interface

- Comes with its own R

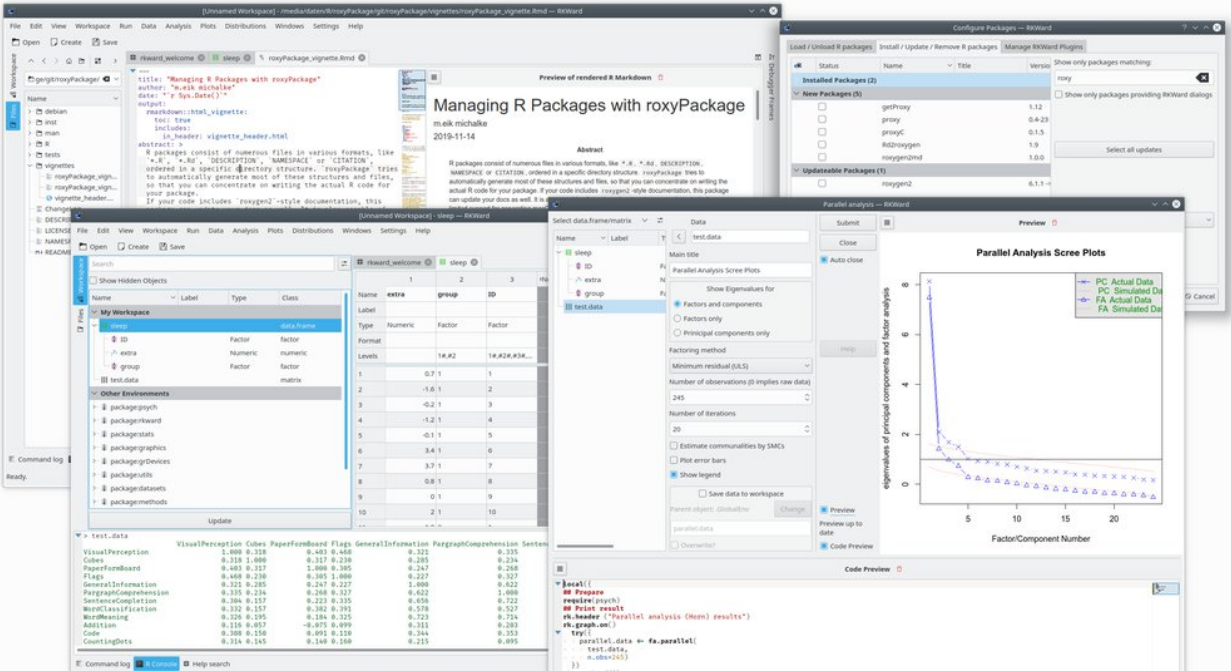
- No need to install it like RStudio
- Can work with your installation

 [News](#) [Download](#) [Documentation](#) [Support](#) [Get Involved](#) Made by KDE

RKward

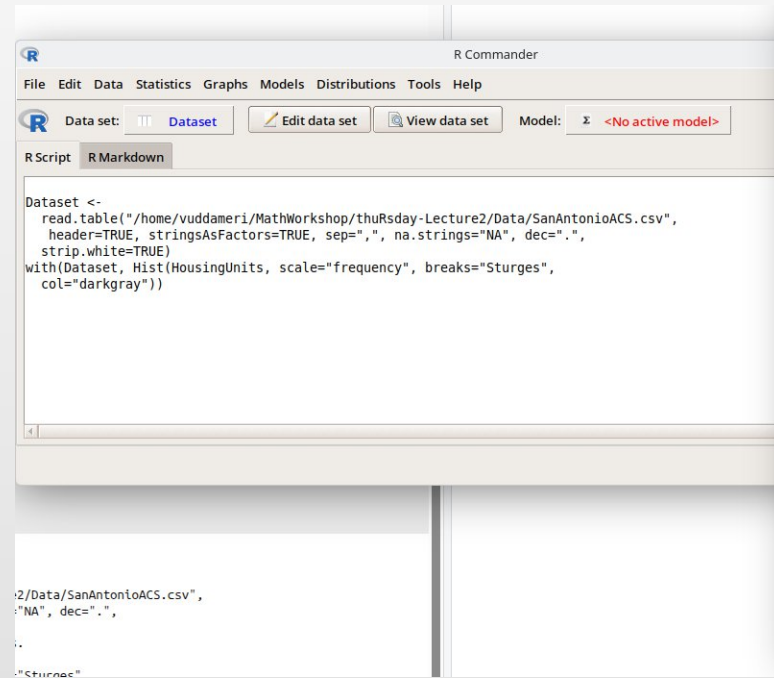
RKward is an easy to use and easily extensible IDE/GUI for R.
It aims to combine the power of the R-language with the ease of use of commercial statistics tools.

[Download RKward 0.8.3](#) - for [macOS](#) | [Windows](#) | [GNU/Linux](#)

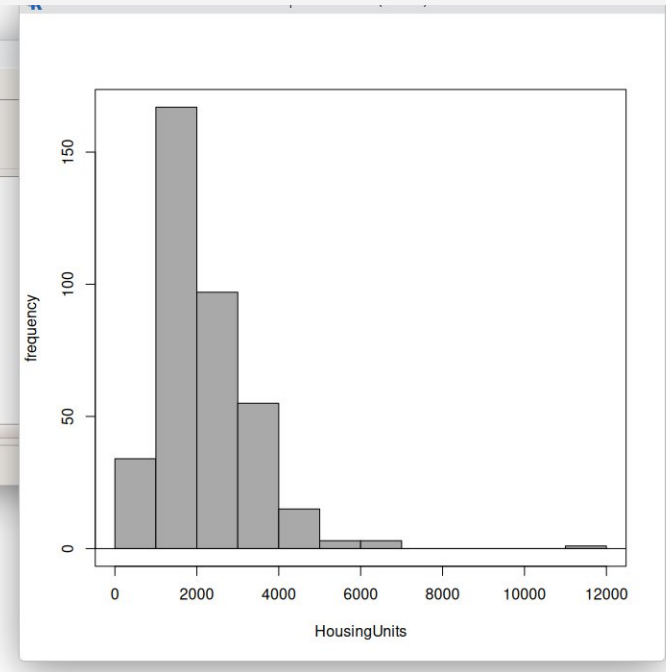


R Commander

- It integrates with R
- You install it as a package
 - `install.packages('Rcmdr')`
- You call the library in R
 - `library(Rcmdr)`
 - Opens a GUI
- Exposes the code it writes
- You can Run from R or RStudio
 - You can add your own code in R or RStudio



```
Dataset <-  
read.table("/home/vuddameri/MathWorkshop/thuRsdai-Lecture2/Data/SanAntonioACS.csv",  
  header=TRUE, stringsAsFactors=TRUE, sep="," na.strings="NA", dec=".",  
  strip.white=TRUE)  
with(Dataset, Hist(HousingUnits, scale="frequency", breaks="Sturges",  
  col="darkgray"))
```



What is the Best Solution?

- Depends upon your goals
- If you want to use R Engine but don't want to learn R
 - Jamovi and RKWard
 - Limited to a small subset of capabilities
 - Mostly statistical
 - Little control on the charts and graphs produced
 - Better than a spreadsheet
- If you want to use R for a broader Data Analytic Framework
 - Statistics, Machine learning, Bayesian Methods, AI
 - Need to invest in learning R Scripting

Hands-On Demonstrations



Two Demos

- Use the San Antonio ACS Data to evaluate whether precincts having populations with Higher Percentage of Bachelors Degrees or Higher have lower poverty rates
 - We can use a regression model and evaluate whether the coefficient is statistically significant
 - Recode variables as factors and perform ANOVA
 - Can solve this both using R Commander (GUI) and via coding
- Visualize the Precipitation amounts at San Antonio Airport using an interactive GIS map
 - Advanced visualization
 - Only possible via scripting